

The Informational Efficiency of Frequency-Report Scoring Rules

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Abstract

In a multinomial frequency-report environment, a subject has latent beliefs p , but the researcher observes only an incentivized count report r . We compare frequency-report scoring rules by how sharply that report identifies the latent beliefs. The inference is recast as a partial-identification problem: each scoring rule maps the report to an identified set of beliefs—those under which the report would have been optimal—from which coordinate and mean bounds follow. The main analytical result is that quadratic-distance frequency scoring gives a transparent projection rule, a linear identified set, closed-form coordinate bounds, and linear-program bounds for means. Discrete-metric scoring is the known fixed-prize rule with closed-form bounds, Manhattan distance has an exact identified set with sharp semi-analytical bounds, and Hamming distance is characterized analytically. A design exercise over the three rules with tractable bounds finds no universal ranking: the most informative rule depends on the inferential objective and the design regime.

1 Introduction

Belief elicitation in experiments often asks subjects to report probabilities directly. Frequency reports ask instead for counts: out of n realizations, how many times will each outcome occur? This format is natural when the object of interest is a distribution of choices, signals, or outcomes in a finite group. It may also be easier for subjects than reporting abstract probabilities under a standard scoring rule. Many probability scoring rules, including the quadratic scoring rule, are well understood theoretically [9, 3, 12], but they require subjects to understand how probability reports map into payments. That mapping can be cognitively demanding, especially when subjects reason more naturally in frequencies than in abstract probabilities [4, 11]. Direct monetary scoring rules also raise the familiar risk-aversion problem [1].

The practical problem is that the researcher observes a count report r , not the subject's latent belief vector p . The report is an incentivized message generated by a scoring rule. Thus the inferential question is inverse: given the rule and the observed report, which beliefs are consistent with optimal reporting? The paper follows the chain

$$S \rightarrow R_S(p) \rightarrow P_S(r) \rightarrow [\underline{p}_{S,i}(r), \bar{p}_{S,i}(r)].$$

Here S is a frequency-report scoring rule, $R_S(p)$ is the set of reports that are optimal for a subject with beliefs p , and

$$P_S(r) = \{p \in \Delta^k : r \in R_S(p)\}$$

is the identified set induced by the observed report. This recasts incentivized belief elicitation as a partial-identification problem [7]: the scoring-rule mechanism, rather than a sampling distribution over repeated observations, supplies the identifying restriction. The final object of the chain is the bound $[\underline{p}_{S,i}(r), \bar{p}_{S,i}(r)]$ it yields on each individual probability p_i . A researcher may also care about functionals of the belief vector, such as a mean or another linear index; bounds on these are obtained from the same identified set $P_S(r)$.

This paper uses the term informational efficiency in an operational sense. A rule is more informative for a given design objective if the observed report leads to narrower coordinate bounds or narrower mean and functional bounds. These inferential gains must be weighed against implementation costs: exact-match payment probabilities, cognitive simplicity, computational burden, and robustness to risk aversion.

The paper’s main analytical contribution is for quadratic-distance frequency scoring. Under this rule, a subject reports the feasible count vector closest to the expected count vector np . This gives a linear identified set and closed-form bounds for each p_i . Mean bounds and other linear functional bounds are then simple linear programs over the same region.

The paper compares this rule to three other count-loss mechanisms along a spectrum of analytical tractability. Schlag–Tremewan frequency guessing is the known discrete-metric exact-match mechanism: it elicits multinomial modes, gives closed-form bounds, and is robust to risk aversion under a fixed-prize implementation [10]. Manhattan-distance scoring has an exact identified set and sharp, semi-analytical coordinate bounds. Hamming-distance scoring has an exact identified set but, for more than two categories, coordinate bounds that resist closed form; it is characterized analytically and is not part of the design comparison.

The contribution is therefore not a new exact-match mechanism and not a general theory of scoring rules; Schlag and Tremewan already derive belief bounds for the frequency-guessing rule. The paper recasts frequency-report belief elicitation as a partial-identification problem—each scoring rule maps an observed report to an identified set of latent beliefs—and uses this lens in two ways. Analytically, it gives a closed-form identified set and coordinate bounds for quadratic-distance scoring, the centerpiece result, and places the other rules on a tractability spectrum from closed form through semi-analytical to computational. In a design exercise over the three rules with tractable bounds, the most informative rule depends on the inferential objective and the design regime, with no universal ranking.

2 Setup

The environment is a finite belief-elicitation problem. There are k possible outcomes, and the subject has latent beliefs

$$p = (p_1, \dots, p_k) \in \Delta^k, \quad p_i \geq 0, \quad \sum_{i=1}^k p_i = 1.$$

The elicitation task asks about $n \in \mathbb{N}$ independent repetitions. If the subject’s beliefs are p , the realized count vector is

$$\omega = (\omega_1, \dots, \omega_k) \sim \text{Mult}(n, p).$$

The set of feasible count vectors is

$$\Omega = \left\{ \omega \in \mathbb{N}_0^k : \sum_{i=1}^k \omega_i = n \right\},$$

and the multinomial probability mass function is

$$\Pr_p(\omega) = \frac{n!}{\omega_1! \cdots \omega_k!} \prod_{i=1}^k p_i^{\omega_i}.$$

The subject reports another feasible count vector

$$r = (r_1, \dots, r_k) \in \Omega.$$

Definition 1 (Frequency-report scoring rule). A frequency-report scoring rule is a function

$$S : \Omega \times \Omega \rightarrow \mathbb{R},$$

where $S(r, \omega)$ is the score assigned to report r when the realized count vector is ω . The paper studies rules that reward reports close to realized counts. A convenient form is

$$S_D(r, \omega) = a - bD(r, \omega), \quad b > 0,$$

where $D : \Omega \times \Omega \rightarrow \mathbb{R}_+$ is a count-error index. The function D need not be a metric: squared distance is a divergence-style count loss, not a metric, because it fails the triangle inequality.

The constants a and b do not matter for expected-score rankings as long as $b > 0$. They matter for implementation. If scores are paid directly as money and nonnegative payments are desired, one can choose

$$a \geq b \max_{r, \omega \in \Omega} D(r, \omega).$$

How the score is implemented as a payment affects robustness to the subject's risk attitude; the paper's discussion of risk aversion and implementation takes up this question separately.

Definition 2 (Optimal reports). Under risk neutrality, the optimal-report correspondence induced by a scoring rule S is

$$R_S(p) = \arg \max_{r \in \Omega} \mathbb{E}_p[S(r, \omega)].$$

Because Ω is finite, $R_S(p)$ is nonempty.

Risk neutrality is the maintained assumption throughout the body of the paper; the discussion of risk aversion and implementation separately shows how the analysis extends to risk-averse expected-utility subjects.

This is the forward incentive object. It answers the question: if the subject's latent beliefs are p , which reports are optimal? The econometric object is the inverse question. After observing r , the researcher asks which beliefs could have generated that report.

Definition 3 (Identified set). For an observed report $r \in \Omega$, the identified set induced by S is

$$P_S(r) = \{p \in \Delta^k : r \in R_S(p)\}.$$

Ties are included: if r is one of several optimal reports under p , then $p \in P_S(r)$.

We call $P_S(r)$ the *mechanism-induced identified set*: the identifying restriction is the optimal-report condition $r \in R_S(p)$ imposed by the scoring-rule mechanism, not a sampling distribution over repeated observations. By construction $P_S(r)$ is exactly the set of beliefs consistent with that restriction.

The most direct summaries of $P_S(r)$ are coordinate-wise identification bounds:

$$\underline{p}_{S,i}(r) = \inf_{p \in P_S(r)} p_i, \quad \bar{p}_{S,i}(r) = \sup_{p \in P_S(r)} p_i,$$

with interval

$$P_{S,i}(r) = [\underline{p}_{S,i}(r), \bar{p}_{S,i}(r)].$$

When the scoring rule is clear from context, the subscript S is omitted.

Many experiments require more than coordinate-by-coordinate probability bounds. If outcomes have numerical values $x_1, \dots, x_k$, the latent mean implied by beliefs p is

$$\mu(p; x) = \sum_{i=1}^k p_i x_i.$$

This could represent an expected outcome, an expected treatment share, a predicted group average, or another linear index of the belief distribution. The sharp mean bounds after observing r are

$$\underline{\mu}_S(r; x) = \inf_{p \in P_S(r)} \sum_{i=1}^k p_i x_i, \quad \bar{\mu}_S(r; x) = \sup_{p \in P_S(r)} \sum_{i=1}^k p_i x_i.$$

The corresponding mean interval is

$$[\underline{\mu}_S(r; x), \bar{\mu}_S(r; x)].$$

These bounds generally require optimizing over the full joint region $P_S(r)$. Coordinate intervals alone do not generally give sharp bounds for a mean, because they ignore dependence across coordinates imposed by the simplex and the scoring rule.

The design comparison later summarizes the informativeness of a rule using coordinate widths,

$$\bar{p}_{S,i}(r) - \underline{p}_{S,i}(r),$$

and functional widths,

$$\bar{\mu}_S(r; x) - \underline{\mu}_S(r; x).$$

Exact-match payment probability is reported only for the discrete-metric rule, as an implementation diagnostic of that fixed-prize mechanism; it is not a cross-rule comparison metric.

The next section applies this setup to the four headline rules: quadratic-distance, discrete-metric, Manhattan, and Hamming.

3 Frequency-Report Scoring Rules

This section applies the setup to four frequency-report scoring rules. They span a spectrum of analytical tractability. Quadratic-distance and discrete-metric scoring have closed-form identified sets and closed-form coordinate bounds. Manhattan-distance scoring has an exact identified set and sharp coordinate bounds in semi-analytical form. Hamming-distance scoring has an exact identified set but, for more than two categories, coordinate bounds that resist closed form. The same inverse-region machinery applies to other count-loss criteria, such as Chebyshev (L_∞) distance, which we do not pursue here.

3.1 Rules with Closed-Form Bounds

3.1.1 Quadratic-Distance Frequency Scoring

The main analytical rule is quadratic-distance frequency scoring. It pays less when the reported count vector is far from the realized count vector in squared Euclidean distance:

$$S_Q(r, \omega) = a - b\|r - \omega\|_2^2, \quad b > 0.$$

This rule is connected in spirit to quadratic scoring rules, but it is not a standard probability score. It is a frequency-report count-loss rule: subjects report counts, and the penalty is the squared difference between reported and realized counts. Write P_Q for the identified set induced by this rule.

Given beliefs p , the expected count vector is np . The key forward fact is that quadratic-distance scoring asks the subject to report the feasible integer count vector closest to np . The inverse question is therefore geometric: for which beliefs is the observed report r the closest feasible count vector to np ?

Proposition 1 (Quadratic distance: reports, regions, and bounds). *For every $p \in \Delta^k$,*

$$R_{S_Q}(p) = \arg \min_{s \in \Omega} \|s - np\|_2^2.$$

For an observed report $r \in \Omega$, the identified set is

$$P_Q(r) = \left\{ p \in \Delta^k : n(p_i - p_j) \leq r_i - r_j + 1 \quad \forall i, j \text{ with } r_j > 0 \right\}.$$

Let

$$m(r) = |\{j : r_j > 0\}|$$

be the number of categories with positive reported counts. The sharp coordinate intervals are

$$P_{Q,i}(r) = \begin{cases} \left[0, \frac{m(r)}{n(m(r) + 1)} \right], & r_i = 0, \\ \left[\frac{r_i - 1}{n} + \frac{1}{nk}, \frac{r_i + 1}{n} - \frac{1}{n m(r)} \right], & r_i > 0. \end{cases}$$

Projection ties are included.

The inequalities compare the observed report r with reports obtained by moving one count from a positive reported category j to another category i . If all such one-count moves weakly increase squared distance from np , then r is a closest feasible report. Thus $P_Q(r)$ is a polytope in the belief simplex. The proof is in Appendix A.3.

Probability and mean bounds. The coordinate bounds are closed form. For numerical outcomes $x_1, \dots, x_k$, sharp mean bounds are obtained by optimizing over the same polytope:

$$\min_{p \in P_Q(r)} \sum_{i=1}^k p_i x_i \quad \text{and} \quad \max_{p \in P_Q(r)} \sum_{i=1}^k p_i x_i.$$

These are linear programs. For coordinate probabilities, the displayed intervals give the corresponding linear-program solutions explicitly. This is the main practical advantage of quadratic-distance frequency scoring: the same transparent region gives both probability bounds and bounds on expected values.

Practical interpretation. Quadratic distance is mean-oriented: it links a report r to beliefs for which r is the integer projection of np . Compared with exact matching, the payment varies smoothly with distance rather than only with exact equality. The main implementation cost is that direct monetary quadratic-distance payments generally require risk neutrality unless the score is implemented probabilistically, as discussed below.

3.1.2 Discrete-Metric Frequency Scoring

The second primary rule is the frequency-guessing mechanism of Schlag and Tremewan [10]. Its count-error index is the count-vector discrete metric,

$$D_0(r, \omega) = \mathbf{1}\{r \neq \omega\},$$

so, in the form $S_D = a - bD$ of Section 2, the score is

$$S_0(r, \omega) = a - b \mathbf{1}\{r \neq \omega\}, \quad b > 0.$$

Equivalently, $S_0(r, \omega) = (a - b) + b \mathbf{1}\{r = \omega\}$: the subject receives an extra b whenever the realized count vector equals the report. This fixed-prize reading is useful for implementation and risk aversion; the two formulations induce the same optimal reports. Write R_0 and P_0 for the report correspondence and identified set induced by this rule.

The mechanism is simple: a subject with beliefs p maximizes the probability of an exact match,

$$\Pr_p(\omega = r) = \frac{n!}{r_1! \cdots r_k!} \prod_{i=1}^k p_i^{r_i}.$$

Thus the forward object is the multinomial mode set. The inverse object is the set of beliefs for which the observed report is one of those modes.

Proposition 2 (Discrete metric: reports, regions, and bounds). *Let $r \in \Omega$, and allow ties in the multinomial mode. Then r is optimal under discrete-metric frequency scoring if and only if r is a multinomial mode:*

$$R_0(p) = \arg \max_{s \in \Omega} \Pr_p(\omega = s).$$

The identified set is

$$P_0(r) = \left\{ p \in \Delta^k : r_j p_i \leq (r_i + 1) p_j \quad \forall i, j \text{ with } r_j > 0 \right\}.$$

For each coordinate, the sharp identification interval is

$$P_{0,i}(r) = \left[\frac{r_i}{n+k-1}, \frac{r_i+1}{n+1} \right].$$

The inequalities say that no one-count transfer from a reported category j to another category i can increase the multinomial probability of the report. Weak inequalities include ties. The formula also handles boundaries: if $r_i = 0$, the lower bound is 0; if $r_i = n$, the upper bound is 1. These are the frequency-guessing bounds in Schlag and Tremewan [10, Proposition 1]; they are restated here to make the informational-efficiency comparison self-contained. The proof is in Appendix A.1.

Probability and mean bounds. The discrete-metric rule gives closed-form coordinate intervals. For linear functionals such as means, sharp bounds use the full joint region, not just the coordinate intervals:

$$\min_{p \in P_0(r)} \sum_{i=1}^k p_i x_i \quad \text{and} \quad \max_{p \in P_0(r)} \sum_{i=1}^k p_i x_i.$$

These are linear programs because $P_0(r)$ is defined by linear inequalities. Thus discrete-metric frequency scoring is analytically convenient for individual probability bounds, while mean inference still requires using the joint mode region.

Practical interpretation. Discrete-metric frequency scoring is transparent and robust to risk aversion because it is a fixed-prize event. Its implementation cost is that the probability of an exact match can be small when n is large or the outcome space is rich. It is a prominent known rule in the comparison, not a new mechanism introduced here.

3.2 Rules Without Closed-Form Bounds

The remaining two headline rules lack closed-form coordinate bounds, but they differ sharply from each other. Manhattan-distance scoring has an exact identified set, and its sharp coordinate bounds reduce to a one-dimensional computation; it is included in the design exercise. Hamming-distance scoring also has an exact identified set, but for more than two categories its sharp bounds resist closed form and are computationally intractable at the (n, k) scales of the design exercise; it is therefore characterized analytically and excluded from the design comparison.

3.2.1 Manhattan-Distance Frequency Scoring

The Manhattan-distance rule is

$$S_M(r, \omega) = a - b \sum_{i=1}^k |r_i - \omega_i|, \quad b > 0.$$

It rewards total absolute count accuracy rather than squared count accuracy. Its inferential interpretation is median-oriented rather than mean-oriented.

Optimal reports and identified set. For category i , the realized count has binomial marginal distribution

$$W_i \sim \text{Bin}(n, p_i).$$

Let

$$\ell_i(t; p_i) = \mathbb{E}|t - W_i|, \quad F_i(t; n, p_i) = \Pr(W_i \leq t).$$

Then optimal reports solve

$$R_M(p) = \arg \min_{r \in \Omega} \sum_{i=1}^k \ell_i(r_i; p_i).$$

The simplex constraint $\sum_i r_i = n$ prevents the researcher from treating each coordinate independently. Because each coordinate loss ℓ_i is discrete-convex and the objective is separable, the absence of any single-count transfer that lowers expected Manhattan loss is both necessary and sufficient for global optimality. Equivalently,

$$P_M(r) = \left\{ p \in \Delta^k : F_i(r_i; n, p_i) \geq F_j(r_j - 1; n, p_j) \quad \forall i, j \text{ with } r_i < n, r_j > 0 \right\}.$$

Weak inequalities include ties; there is no receiver constraint when $r_i = n$ and no sender constraint when $r_j = 0$. Single-unit-transfer optimality is sufficient, not merely necessary, so $P_M(r)$ is the exact identified set rather than an outer approximation. The proof is in Appendix A.2.

Identification bounds. For $k = 2$, write $r = (t, n - t)$. Then Manhattan loss is $2|t - W|$, with $W \sim \text{Bin}(n, p_1)$, so the report identifies the binomial-median interval

$$P_{M,1}(r) = \{p_1 \in [0, 1] : F(t - 1; n, p_1) \leq 1/2 \leq F(t; n, p_1)\}.$$

This binary case is clean.

For $k > 2$, the pairwise CDF inequalities are equivalent to a single threshold condition: $p \in P_M(r)$ if and only if there exists $c \in [0, 1]$ such that

$$F_i(r_i - 1; n, p_i) \leq c \leq F_i(r_i; n, p_i) \quad \forall i,$$

using $F_i(-1; n, p_i) = 0$ and $F_i(n; n, p_i) = 1$. For a fixed c , these inequalities give a coordinate box through inverse binomial CDFs, and intersecting that box with the simplex gives the feasible beliefs at that threshold. Each sharp coordinate bound then reduces to a one-dimensional optimization over the threshold c : the objective is unimodal in c , so the bound is the solution of a single monotone scalar equation. The coordinate bounds are thus sharp and semi-analytical—not closed form, because the inverse binomial CDF is not elementary, but exact and one-dimensional rather than grid-limited. The sharp mean bounds are obtained by a one-dimensional search over the same threshold.

Practical interpretation. Manhattan distance is useful when a median-type count prediction is substantively appropriate. It is also useful as a contrast: changing the distance criterion changes the inferred identified set. The multi-category Manhattan bounds are sharp and reduce to a one-dimensional threshold computation; they are less transparent than the closed-form quadratic-distance bounds but are exact, not approximations.

3.2.2 Hamming-Distance Frequency Scoring

The Hamming-distance rule penalizes the number of categories whose reported count differs from the realized count:

$$S_H(r, \omega) = a - b \sum_{i=1}^k \mathbf{1}\{r_i \neq \omega_i\}, \quad b > 0.$$

Its expected loss separates into binomial marginal exact-coordinate probabilities:

$$\mathbb{E}_p \left[\sum_i \mathbf{1}\{r_i \neq \omega_i\} \right] = \sum_i \{1 - \Pr(\text{Bin}(n, p_i) = r_i)\}.$$

Thus the forward objective is computationally simple, even though the reported counts still have to sum to n .

For an observed report r , define

$$L_H(r; p) = \sum_i \{1 - \Pr(\text{Bin}(n, p_i) = r_i)\}.$$

The identified set is exactly

$$P_H(r) = \{p \in \Delta^k : L_H(r; p) \leq L_H(s; p) \ \forall s \in \Omega\}.$$

This finite set of expected-loss inequalities fully specifies identification.

For $k = 2$, Hamming distance coincides with discrete-metric frequency scoring: matching one coordinate is equivalent to matching the whole binary count vector, so the report incentives are identical, and $P_H(r)$ inherits the closed-form discrete-metric coordinate interval.

For $k > 2$ the rule is markedly harder. Each coordinate term $\Pr(\text{Bin}(n, p_i) = r_i)$ is unimodal in the reported count but not discrete-concave, so a single-unit-transfer argument does not establish global optimality: a multi-count transfer can move the report from one binomial mode to a higher one, and single-unit-transfer optimality is necessary but not sufficient. The identified set $P_H(r)$ is therefore genuinely the system above—a non-convex semialgebraic set, not a transfer polytope. A two-sided sandwich is nonetheless available. The modal box $\prod_i [r_i/(n+1), (r_i+1)/(n+1)]$, the set of beliefs at which every r_i is a mode of its binomial marginal, is a closed-form inner bound: when every coordinate sits at its marginal mode, r maximizes the separable objective term by term. The single-unit-transfer inequalities give an outer bound. The sharp coordinate bounds lie between these and are obtained only by finite numerical computation; at the (n, k) scales of the design exercise this computation is intractable, so Hamming distance is characterized analytically here and excluded from the design comparison.

4 Informational Efficiency: A Design Comparison

The analytical regions are useful because they tell an experimenter what can be inferred after observing a report. This section reports a design comparison generated by the reproducible script `scripts/design_efficiency.py`. The exercise draws latent beliefs

$$p \sim \text{Dirichlet}(\alpha, \dots, \alpha)$$

and, for each of the three rules with tractable sharp bounds—discrete-metric, quadratic-distance, and Manhattan distance—computes an optimal report $r_S(p)$ and the bounds induced by $P_S(r_S(p))$. Hamming distance is characterized analytically in the preceding section and is not part of this comparison: its sharp identified set is computationally intractable at these (n, k) scales. The final run uses 5,000 belief draws for each cell in

$$n \in \{5, 10, 20, 50\}, \quad k \in \{2, 3, 5, 10\}, \quad \alpha \in \{0.1, 0.3, 1, 3, 10\}.$$

The Dirichlet parameter α sets belief concentration: small α draws sparse beliefs near a vertex of the simplex, large α draws balanced, interior beliefs. For each rule the comparison records two inferential targets—the average width of the coordinate bounds, and the width of the bound on the linear functional $\sum_i p_i x_i$ with outcome values $x_i = (i - 1)/(k - 1)$. Manhattan coordinate bounds are sharp, computed by a threshold root-find; Manhattan mean bounds are threshold-computed with tolerance 10^{-4} .

The comparison is conditional on the theoretical characterizations above. It is not evidence of incentive compatibility, and it is not an optimality theorem. It is a diagnostic for the practical question: which rule gives a researcher tighter inference for the object of interest?

The ranking is governed by belief concentration. For the two rules with closed-form bounds the crossover can be read from those bounds before any simulation. The discrete-metric coordinate width, $[r_i(k - 2) + (n + k - 1)]/[(n + 1)(n + k - 1)]$, is linear in the reported count r_i ; because $\sum_i r_i = n$, its average over the k coordinates is a constant, $(k - 1)(2n + k)/[k(n + 1)(n + k - 1)]$, the same for every report. The quadratic-distance coordinate width for a positive-report category, $2/n - 1/(nm) - 1/(nk)$ with $m = m(r)$ the number of categories with positive reported counts, does not depend on r_i and shrinks as the report concentrates on fewer categories. A report concentrated on few categories therefore has a small average quadratic width; a balanced report, with $m = k$, has an average quadratic width that exceeds the discrete-metric constant. The ranking of the two closed-form rules thus crosses over with belief concentration as a matter of the closed forms alone. The design comparison confirms and quantifies this crossover, and extends it to Manhattan distance, which has no closed-form bound.

Averaged over the whole grid the three rules barely differ—mean coordinate width 0.103, 0.104, and 0.102 for discrete-metric, quadratic-distance, and Manhattan scoring—because each rule’s favourable and unfavourable regimes cancel. Table 1 reports the win shares—the share of belief draws on which a rule gives the narrowest interval—broken out by α .

α	Average-coordinate			Linear-mean		
	Discrete	Quadratic	Manhattan	Discrete	Quadratic	Manhattan
0.1	0.08	0.84	0.07	0.08	0.83	0.10
0.3	0.28	0.58	0.14	0.22	0.60	0.18
1	0.76	0.16	0.08	0.58	0.27	0.15
3	0.98	0.01	0.01	0.83	0.13	0.04
10	1.00	0.00	0.00	0.92	0.06	0.02

Table 1: Win share by belief concentration α : the share of belief draws on which a rule gives the narrowest interval, averaged over all n and k at each α . Quadratic-distance scoring wins for sparse beliefs (small α); discrete-metric scoring for balanced beliefs (large α).

Quadratic-distance scoring is the most informative rule when beliefs are sparse and discrete-metric scoring when they are balanced, with a crossover near $\alpha = 1$. At $\alpha = 0.1$ quadratic gives the narrowest average-coordinate interval on 0.84 of draws; by $\alpha \geq 3$ discrete-metric does so on essentially every draw. The linear-mean target shows the same crossover. The number of categories k and the number of trials n are second-order, as Table 3 in Appendix B documents: for the coordinate target the ranking is nearly flat in both, while for the mean target larger n mildly favours quadratic and the binary case $k = 2$ favours discrete-metric. The worst-coordinate target follows the same pattern shifted in quadratic’s favour—quadratic gives the narrowest maximum-coordinate interval on a majority of draws for every $\alpha \leq 3$.

The crossover has a mechanism in the projection interpretation of quadratic-distance scoring. A subject reports the integer count vector closest to np ; when beliefs are sparse, np lies near a vertex of the simplex, the feasible reports around it are extreme count vectors, and the projection polytope is tight, so the induced bounds are narrow. As beliefs become balanced, np moves into the interior, the projection polytope widens, and discrete-metric scoring—which elicits the multinomial mode—becomes the more informative rule.

The cost of the wrong rule, and Manhattan as a hedge. Win shares count only first places. Cell-best regret—a rule’s interval width minus the narrowest width available in the same cell—measures how much precision is forfeited by not using the best rule. Table 2 reports regret by α .

α	Average-coordinate			Linear-mean		
	Discrete	Quadratic	Manhattan	Discrete	Quadratic	Manhattan
0.1	0.032	0.002	0.011	0.047	0.002	0.013
0.3	0.016	0.005	0.007	0.024	0.005	0.009
1	0.003	0.012	0.007	0.006	0.013	0.008
3	0.000	0.019	0.011	0.001	0.020	0.012
10	0.000	0.023	0.014	0.000	0.024	0.016

Table 2: Mean cell-best regret by belief concentration α : a rule’s interval width minus the smallest width among the three rules, averaged over all n and k at each α . Lower is better. Discrete-metric and quadratic scoring each swing between near-zero and substantial regret; Manhattan’s regret is nearly flat.

Discrete-metric and quadratic-distance scoring each have a regime of near-zero regret and a regime in which regret is substantial. Discrete-metric’s average-coordinate regret runs from 0.000 at $\alpha = 10$ to 0.032 at $\alpha = 0.1$; quadratic’s runs the other way, reaching 0.024 on the mean target at $\alpha = 10$. Against typical interval widths near 0.10 to 0.15, the worst-regime values are a quarter or more of an interval width—a real loss of precision from choosing the rule that does not match the belief regime. Manhattan distance behaves differently. It is never the narrowest rule, but its regret is nearly constant in α —between 0.007 and 0.016 at every concentration and for both targets—so it is never far from the best rule either. Manhattan is therefore the regime-robust choice: a researcher who cannot anticipate how concentrated subjects’ beliefs are minimises worst-case regret by using it, whereas committing in advance to discrete-metric or quadratic scoring risks the unfavourable regime. This is the precise sense in which Manhattan is a low-regret rule: it hedges the belief-concentration uncertainty to which the other two rules are exposed.

Payment probability. The discrete-metric rule has a fixed-prize implementation, which is attractive under risk aversion. Its implementation cost is that the exact-match payment probability can be small. In the final run, the average exact-match probability is high for very sparse binary beliefs, but it falls sharply as n , k , and belief balance increase; for instance, it is essentially zero in the $n = 50, k = 10, \alpha \in \{1, 3, 10\}$ cells. Thus narrow bounds under discrete-metric scoring should be weighed against the chance that the subject rarely wins the fixed prize.

5 Discussion

The paper studies frequency-report scoring rules as inferential devices. The report r is not the belief vector. It is evidence about p , and the scoring rule determines how informative that evidence is. The relevant chain is therefore

$$S \rightarrow R_S(p) \rightarrow P_S(r) \rightarrow [\underline{p}_{S,i}(r), \bar{p}_{S,i}(r)].$$

The design comparison shows that the most informative rule depends on how concentrated the subject’s beliefs are; three practical recommendations follow. First, use discrete-metric frequency scoring when the subject is expected to spread probability fairly evenly across the outcomes, with no single outcome carrying most of the probability. It is the most informative rule in that regime, has closed-form coordinate bounds, and is the only rule robust to risk aversion under direct payment; its implementation cost is that exact-match payment probabilities can be low away from concentrated reports.

Second, use quadratic-distance frequency scoring when the subject is expected to place most of the probability on one or a few outcomes. It is the most informative rule in that regime—for the average-coordinate, worst-coordinate, and mean targets alike—and it offers a transparent projection rule, a linear identified set, closed-form coordinate bounds, and linear-program bounds for any linear functional.

Third, use Manhattan distance when it cannot be anticipated in advance whether the subject’s beliefs will be concentrated on a few outcomes or spread across many. Its identified set is exact and its coordinate bounds are sharp—semi-analytical, reducing to a one-dimensional threshold computation rather than a closed form. Manhattan rarely gives the single narrowest interval, but the design comparison shows its regret is nearly constant across belief concentrations: it is never far from the best rule, and so minimizes worst-case regret for a researcher who does not know the regime. Its inferential interpretation is also median-oriented, which suits settings where a median-type count prediction is substantively appropriate.

5.1 Risk Aversion and Implementation

The body of the paper assumes risk neutrality. Whether that assumption can be relaxed depends on how the scoring rule is implemented as a payment. The distinction is between paying the score directly as money and using the score to determine the probability of a fixed prize.

Direct monetary scores. If the subject is paid the score itself, the subject solves $\arg \max_{r \in \Omega} \mathbb{E}_p[u(S(r, \omega))]$. For the discrete-metric rule this does not change incentives. The payment is a fixed prize B if

$r = \omega$ and zero otherwise, so

$$\mathbb{E}_p[u] = u(0) + \Pr_p(\omega = r)\{u(B) - u(0)\},$$

and any strictly increasing u preserves the same optimal reports as maximizing the exact-match probability. Direct monetary distance scores do not have this robustness: quadratic and Manhattan distance generate several possible payment levels, so nonlinear utility can change the ranking of reports. This is the standard risk-aversion problem for monetary scoring rules [1, 11].

Probability implementation. The distance rules can instead be implemented as a binary lottery. Let $q(r, \omega) \in [0, 1]$ be the score normalized into a probability, and pay a fixed prize B with probability $q(r, \omega)$ and zero otherwise. If the prize and outside payment are fixed, utility depends only on final money, and the randomization is objective and independent, then

$$\mathbb{E}_p[q(r, \omega)u(B) + (1 - q(r, \omega))u(0)] = u(0) + (u(B) - u(0)) \mathbb{E}_p[q(r, \omega)],$$

so maximizing expected utility is equivalent to maximizing the expected normalized score, regardless of the curvature of u . A positive affine normalization of the bounded score into $[0, 1]$ preserves the expected-score ranking of reports, so the body’s risk-neutral analysis carries over unchanged. This is the binary-lottery, or binarized-scoring-rule, technique [8, 2, 6, 5]; the contribution here is to apply it, not to introduce it.

Two qualifications matter. First, the robustness is to risk aversion within expected utility. Subjects who depart from expected utility—for instance through probability weighting—can report non-truthfully even under the binary-lottery implementation, and the paper does not address such departures. Second, the binary-lottery implementation asks subjects to reason about a compound lottery on top of the count report, which partially offsets the cognitive simplicity that motivates frequency reports in the first place.

The implementation choice also interacts with a cost noted in the design exercise. The discrete-metric prize is paid only on an exact match, and the exact-match probability falls sharply as n , k , and belief balance increase, so at larger designs the rule becomes a lottery with a vanishing chance of any payment. The discrete-metric rule is thus robust to risk aversion even under direct payment but can become nearly unwinable; the distance rules avoid that under the binary-lottery implementation, whose payment probability is the expected normalized score, at the cost of relying on expected utility.

The paper should not be read as a general theory of discrete scoring rules. Hamming distance is characterized analytically rather than through the design exercise, and other count-loss criteria such as Chebyshev distance are noted only in passing. The intended contribution is narrower: a partial-identification account of frequency reports—turning an observed report into an identified set and finite-sample bounds—together with a design comparison of the rules for which those bounds are tractable.

Important open questions remain. The design exercise should be refined before empirical use, especially with alternative outcome vectors and robustness checks for tie-breaking conventions. The cognitive advantage of frequency reports and distance payments should be tested experimentally rather than assumed. Those extensions are separate from the main point: different frequency-report scoring rules induce different identified sets, and those sets determine the informational efficiency of the mechanism.

A Proofs

A.1 Discrete-Metric Frequency Scoring

Proof of Proposition 2. For any report r ,

$$\mathbb{E}_p[S_0(r, \omega)] = a - b \Pr_p(\omega \neq r) = (a - b) + b \Pr_p(\omega = r),$$

so, since $b > 0$, expected-score maximization is equivalent to maximizing the multinomial probability mass $\Pr_p(\omega = r)$.

It remains to characterize the mode region. Compare r with $r + e_i - e_j$, which is feasible when $r_j > 0$. If $p_j = 0$ and $r_j > 0$, then r cannot be a mode unless all probability is zero, which is impossible. For $p_j > 0$,

$$\frac{\Pr_p(r + e_i - e_j)}{\Pr_p(r)} = \frac{r_j}{r_i + 1} \frac{p_i}{p_j}.$$

Thus a necessary condition for r to be a mode is

$$r_j p_i \leq (r_i + 1) p_j \quad \forall i, j \text{ with } r_j > 0.$$

Conversely, if these inequalities hold, any feasible s can be reached from r by a sequence of one-count transfers from coordinates where the current vector exceeds s to coordinates where it is below s . At each step the same ratio is at most one, because source counts weakly decrease and destination counts weakly increase along the path. Therefore no feasible s has larger multinomial mass than r .

The coordinate upper bound follows by summing

$$r_j p_i \leq (r_i + 1) p_j$$

over $j \neq i$:

$$(n - r_i) p_i \leq (r_i + 1)(1 - p_i),$$

so $p_i \leq (r_i + 1)/(n + 1)$. The lower bound follows by summing

$$r_i p_j \leq (r_j + 1) p_i$$

over $j \neq i$:

$$r_i(1 - p_i) \leq (n - r_i + k - 1) p_i,$$

so $p_i \geq r_i/(n + k - 1)$.

The bounds are sharp. The upper endpoint is attained by

$$p_i = \frac{r_i + 1}{n + 1}, \quad p_j = \frac{r_j}{n + 1} \quad (j \neq i).$$

The lower endpoint, when $r_i > 0$, is attained by

$$p_i = \frac{r_i}{n + k - 1}, \quad p_j = \frac{r_j + 1}{n + k - 1} \quad (j \neq i).$$

When $r_i = 0$, the lower endpoint 0 is attained by setting $p_i = 0$ and assigning probabilities proportional to r_j on the reported support. When $r_i = n$, the upper endpoint 1 is attained by $p_i = 1$. $\square$

A.2 Manhattan-Distance Rule

The Manhattan characterization used in Section 3.2 follows from a standard exchange argument. Linearity of expectation gives

$$\mathbb{E}_p \left[\sum_{i=1}^k |r_i - \omega_i| \right] = \sum_{i=1}^k \mathbb{E}_p |r_i - \omega_i|.$$

The i -th term depends only on the binomial marginal $W_i \sim \text{Bin}(n, p_i)$. For

$$\ell_i(t; p_i) = \mathbb{E}|t - W_i|,$$

the marginal cost of increasing the reported count from t to $t + 1$ is

$$\ell_i(t + 1; p_i) - \ell_i(t; p_i) = 2F_i(t; n, p_i) - 1.$$

Moving one reported count from j to i changes expected Manhattan loss by

$$\{2F_i(r_i; n, p_i) - 1\} - \{2F_j(r_j - 1; n, p_j) - 1\}.$$

Hence no feasible one-count transfer improves the report if and only if

$$F_i(r_i; n, p_i) \geq F_j(r_j - 1; n, p_j) \quad \forall i, j \text{ with } r_i < n, r_j > 0.$$

This local no-transfer condition is also sufficient for global optimality. Each coordinate loss ℓ_i is discrete-convex—its increments $2F_i(t; n, p_i) - 1$ are weakly increasing in t —and the objective $\sum_i \ell_i(r_i; p_i)$ is minimized over the integer simplex $\{r : \sum_i r_i = n\}$. Suppose r satisfies the no-transfer condition and let s be any feasible report. The difference $s - r$ sums to zero, so s is reached from r by a sequence of single-count transfers; choose the sequence monotone, at each step moving a unit from a coordinate currently above its target s to one currently below. A coordinate below its target only ever receives counts and so remains below it, and a coordinate above its target only ever sends counts and remains above it; hence each coordinate's count moves monotonically between r_i and s_i , and every intermediate report is feasible. At a step that moves a unit from j to i , the receiving count c_i satisfies $c_i \geq r_i$ and the sending count c_j satisfies $c_j \leq r_j$; by discrete-convexity the step's cost $\{\ell_i(c_i + 1) - \ell_i(c_i)\} + \{\ell_j(c_j - 1) - \ell_j(c_j)\}$ is at least the cost of the corresponding transfer evaluated at r , which is nonnegative because r satisfies the no-transfer condition. Summing over the path gives $\sum_i \ell_i(s_i) \geq \sum_i \ell_i(r_i)$, so r is globally optimal. Hence $P_M(r)$ is the exact identified set.

The threshold representation follows by taking c between

$$\max_{j: r_j > 0} F_j(r_j - 1; n, p_j) \quad \text{and} \quad \min_{i: r_i < n} F_i(r_i; n, p_i).$$

Inverting the resulting binomial-CDF inequalities coordinate by coordinate gives the threshold-indexed box-simplex description used for Manhattan bounds.

A.3 Quadratic-Distance Rule

Proof of Proposition 1. Maximizing S_2 is equivalent to minimizing $\mathbb{E}_p \|r - \omega\|_2^2$. For fixed r ,

$$\mathbb{E}_p \|r - \omega\|_2^2 = \|r - \mathbb{E}_p[\omega]\|_2^2 + \sum_{i=1}^k \text{Var}_p(\omega_i).$$

The variance term is independent of r , and $\mathbb{E}_p[\omega] = np$. Hence optimal reports are exactly the feasible integer projections of np .

The full inverse projection condition is

$$\|r - np\|_2^2 \leq \|s - np\|_2^2 \quad \forall s \in \Omega,$$

or equivalently

$$2np \cdot (s - r) \leq \|s\|_2^2 - \|r\|_2^2.$$

Taking $s = r + e_i - e_j$, feasible when $r_j > 0$, gives

$$n(p_i - p_j) \leq r_i - r_j + 1.$$

Conversely, suppose all these one-count inequalities hold. For any $s \in \Omega$, write $d = s - r$. Pair each positive unit of d_i with a negative unit of d_j . Each coordinate j with $d_j < 0$ has $s_j < r_j$ and hence $r_j > 0$, so the one-count inequality $n(p_i - p_j) \leq r_i - r_j + 1$ is available for every paired transfer. Summing the one-count inequalities over the paired transfers gives

$$np \cdot d \leq r \cdot d + M,$$

where $M = \sum_{i: d_i > 0} d_i$ is the number of transferred counts. Because d has integer components, $|d_i| \geq 1$ on its support, so

$$\sum_i d_i^2 \geq \sum_i |d_i| = 2M.$$

Doubling the displayed inequality and substituting $2M \leq \sum_i d_i^2 = \|d\|_2^2$ gives $2np \cdot d \leq 2r \cdot d + \|d\|_2^2$. Since $d = s - r$,

$$2np \cdot (s - r) \leq 2r \cdot (s - r) + \|s - r\|_2^2 = \|s\|_2^2 - \|r\|_2^2,$$

which is the full projection condition.

It remains to derive the coordinate bounds. Let $m = m(r) = |\{j : r_j > 0\}|$. If $r_i > 0$, summing

$$n(p_u - p_i) \leq r_u - r_i + 1$$

over all $u \neq i$ gives

$$1 - p_i \leq (k - 1)p_i + \frac{n - kr_i + k - 1}{n},$$

so

$$p_i \geq \frac{r_i - 1}{n} + \frac{1}{nk}.$$

For the upper bound, sum

$$n(p_i - p_v) \leq r_i - r_v + 1$$

over positive-report coordinates $v \neq i$, and use nonnegativity of the zero-report coordinates:

$$1 - p_i \geq (m - 1)p_i + \frac{n - mr_i - m + 1}{n}.$$

This gives

$$p_i \leq \frac{r_i + 1}{n} - \frac{1}{nm}.$$

If $r_i = 0$, the lower bound is 0. For the upper bound, sum

$$n(p_i - p_v) \leq 1 - r_v$$

over all m positive-report coordinates v :

$$1 - p_i \geq mp_i + \frac{n - m}{n},$$

so

$$p_i \leq \frac{m}{n(m + 1)}.$$

The bounds are sharp, with an explicit attaining belief for each endpoint. For the lower bound with $r_i > 0$, take $p_i = (r_i - 1)/n + 1/(nk)$ and $p_u = r_u/n + 1/(nk)$ for $u \neq i$. This is a probability vector—it sums to one and every coordinate is at least $1/(nk)$ —and $n(p_a - p_b) = r_a - r_b \leq r_a - r_b + 1$ for $a, b \neq i$ while $n(p_a - p_i) = r_a - r_i + 1$, so it lies in $P_Q(r)$ and the constraints with $b = i$ bind, attaining the lower endpoint. For the upper bound with $r_i > 0$, take $p_i = (r_i + 1)/n - 1/(nm)$, take $p_v = r_v/n - 1/(nm)$ for the positive-report coordinates $v \neq i$, and set $p_v = 0$ for the zero-report coordinates. This sums to one and is nonnegative, lies in $P_Q(r)$, and makes the constraints $n(p_i - p_v) \leq r_i - r_v + 1$ bind for every positive-report $v \neq i$, attaining the upper endpoint. For the upper bound with $r_i = 0$, take $p_i = m/(n(m + 1))$, take $p_v = \{(m + 1)r_v - 1\}/(n(m + 1))$ for the positive-report coordinates v , and set $p_w = 0$ for the other zero-report coordinates. This sums to one and is nonnegative, lies in $P_Q(r)$, and makes the constraints $n(p_i - p_v) \leq 1 - r_v$ bind, attaining the upper endpoint. Thus the displayed bounds are sharp. $\square$

B Design Exercise Details

The design exercise is generated by the script `scripts/design_efficiency.py`. The final grid uses $n \in \{5, 10, 20, 50\}$, $k \in \{2, 3, 5, 10\}$, $\alpha \in \{0.1, 0.3, 1, 3, 10\}$, and 5,000 draws from the symmetric Dirichlet distribution for each cell. For each draw, and for each of the three rules in the comparison—discrete-metric, quadratic-distance, and Manhattan—the script computes an optimal report and then bounds for the identified set generated by that report. Hamming distance is excluded: its sharp bounds are computationally intractable at this grid’s scale.

Discrete-metric coordinate bounds are closed form and mean bounds are linear programs over the transfer region. Quadratic-distance coordinate bounds are closed form and mean bounds are linear programs over the one-count transfer polytope. Manhattan coordinate bounds are sharp: the script solves the one-dimensional threshold optimization implied by Appendix A.2 with a monotone root-find. Manhattan mean bounds use the same threshold representation but are computed over a threshold grid with tolerance 10^{-4} .

The generated files are CSV tables under `outputs/design_exercise/`. These outputs are diagnostics for experimental design. They are not simulations of subject behavior and do not replace the mathematical characterization of $R_S(p)$ and $P_S(r)$.

Table 3 substantiates the claim in Section 4 that the number of categories k and the number of trials n are second-order. It reports win shares marginally by k and by n , averaging over the other two design dimensions. Against the variation across belief concentration α in Table 1—where the leading rule’s win share runs from near zero to one—the dependence on k and n is slight.

	Average-coordinate			Linear-mean		
	Discrete	Quadratic	Manhattan	Discrete	Quadratic	Manhattan
By number of categories k						
2	0.72	0.25	0.03	0.72	0.25	0.03
3	0.59	0.37	0.04	0.34	0.51	0.15
5	0.58	0.35	0.07	0.49	0.42	0.09
10	0.59	0.31	0.10	0.56	0.33	0.11
By number of trials n						
5	0.60	0.34	0.06	0.57	0.33	0.10
10	0.61	0.32	0.07	0.57	0.33	0.11
20	0.62	0.31	0.06	0.53	0.38	0.09
50	0.65	0.30	0.05	0.44	0.47	0.09

Table 3: Win share marginally by number of categories k and number of trials n , averaging over the other two design dimensions. Relative to belief concentration (Table 1), the dependence is second-order: the coordinate target is nearly flat in both, and the only visible movements are the binary case $k = 2$ and a mild drift toward quadratic-distance scoring at large n for the mean target.

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